WATERWATCH · TRACEABILITY MATRIX

WaterWatch traceability matrix

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	WWT-RTM-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
Source	docs/03b-traceability.md @ 59e5426		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

WaterWatch traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 16 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-009	PR-010	Caretaker maintenance	WWT-CAL-001, WWT-STM-001	Task analysis (WWT-CAL-001 J3), TRL 3; timed visits, TRL 5	At risk
UR-002	PR-002	Turbidity	WWT-CAL-001, WWT-BLD-001, WWT-STM-001	Bubble and settling calculation (WWT-CAL-001 F), TRL 3; stabilised standards, TRL 4	At risk
UR-006	PR-009	Outdoor survival	WWT-CAL-001, WWT-BLD-001	Thermal estimate (WWT-CAL-001 C), TRL 3; logged enclosure temperature, TRL 4	At risk
UR-011	PR-012	Parts cost	WWT-CAL-001	Priced BOM (WWT-CAL-001 K1), TRL 3	Over target
UR-003	PR-005	Alerts	WWT-CAL-001, WWT-BLD-001, WWT-STM-001	Latency calculation (WWT-CAL-001 H), TRL 3; network test, TRL 4	Met on paper
UR-004	PR-006	Data	WWT-CAL-001, WWT-BLD-001	Data budget (WWT-CAL-001 G), TRL 3	Met on paper
UR-005	PR-007	Solar autonomy	WWT-CAL-001, WWT-BLD-001	Energy budget (WWT-CAL-001 A, B), TRL 3; logged energy, TRL 4	Met on paper
UR-001	PR-001	Free chlorine	WWT-CAL-001, WWT-BLD-001, WWT-STM-001, WWT-OPT-001	Analysis (WWT-CAL-001 E), TRL 3; side by side with the shared DPD photometer (O2), TRL 4	Met on paper: 8 of 8 cases with the photometer as the monthly reference, worst ±0.411 against ±0.450 mg/L (WWT-CAL-001 E12)
UR-008	PR-008	Water use	WWT-CAL-001, WWT-BLD-001, WWT-STM-001	Flow calculation (WWT-CAL-001 D), TRL 3; measured, TRL 4	Met on paper
UR-010, UR-007	PR-011	Installation	WWT-CAL-001, WWT-BLD-001	Task analysis (WWT-CAL-001 J1, J2), TRL 3; timed install, TRL 5	Met on paper
UR-013	PR-003	Water temperature	WWT-CAL-001, WWT-BLD-001	Inspection of the datasheet, TRL 3	Met by design
UR-016	PR-004	Schedule	WWT-CAL-001, WWT-BLD-001, WWT-STM-001	Analysis, TRL 3; firmware demonstration, TRL 4	Met by design
UR-012	PR-013	Measurements, not verdicts	WWT-CAL-001, WWT-BLD-001, WWT-STM-001	Inspection of the alert templates (WWT-DEC-001), TRL 3	Met by design

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-007	PR-014	Supply protection	WWT-BLD-001, WWT-STM-001	Inspection of the drawing WWT-DWG-001 and BOM, TRL 3; backflow test, TRL 4	Met by design
UR-014	PR-015	Open files	WWT-STM-001	Inspection of the repository and REUSE lint, TRL 3	Met by design
UR-015	PR-016	Battery and chemical handling	WWT-STM-001	Inspection of the BOM and build plan, TRL 3; charger test, TRL 4	Met by design

Gaps

None.